Creating and Destroying Breathable Oxygen

The oxygen we breathe is O_2: a molecule made of two oxygen atoms, bound together by electrons. There is lots of oxygen on Earth in other forms: carbon dioxide, water, minerals in the Earth's crust, to name a few. But our bodies can't use that oxygen until some organism liberates it and converts it to O_2.

The atmosphere's O_2 is *destroyed* by breathing, burning, and decay. When we breathe in O_2 our bodies combine it with carbon to form carbon dioxide, CO_2, releasing lots of energy that our bodies use. When wood is burned, the flames rapidly combine the atmosphere's O_2 with the wood's carbon to form CO_2, which generates the heat that keeps the burning going. When dead plants decay on the forest floor, their carbon is slowly combined with the atmosphere's O_2 to form CO_2 and heat.

The atmosphere's O_2 is *created* primarily by photosynthesis: chloroplasts in plants[24] (Chapter 11) use energy from sunlight to split CO_2 into C and O_2. The O_2 is liberated into the Earth's atmosphere, while the plants combine the carbon with hydrogen and oxygen from water to form the carbohydrates that they need for growth.

O_2 Destruction and CO_2 Poisoning

Suppose evolution creates a pathogen that destroys chloroplasts, as speculated by Elliot Meyerowitz at the end of the last chapter. Photosynthesis ends, not all at once, but gradually as plants die out. O_2 is no longer being created, but it is still being destroyed by breathing, burning, and decay—primarily decay, it turns out. Fortunately for the remaining humans, there is not enough decaying plant life on the Earth's surface to swallow up all the O_2.